Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 24 GB of memory. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy during training.

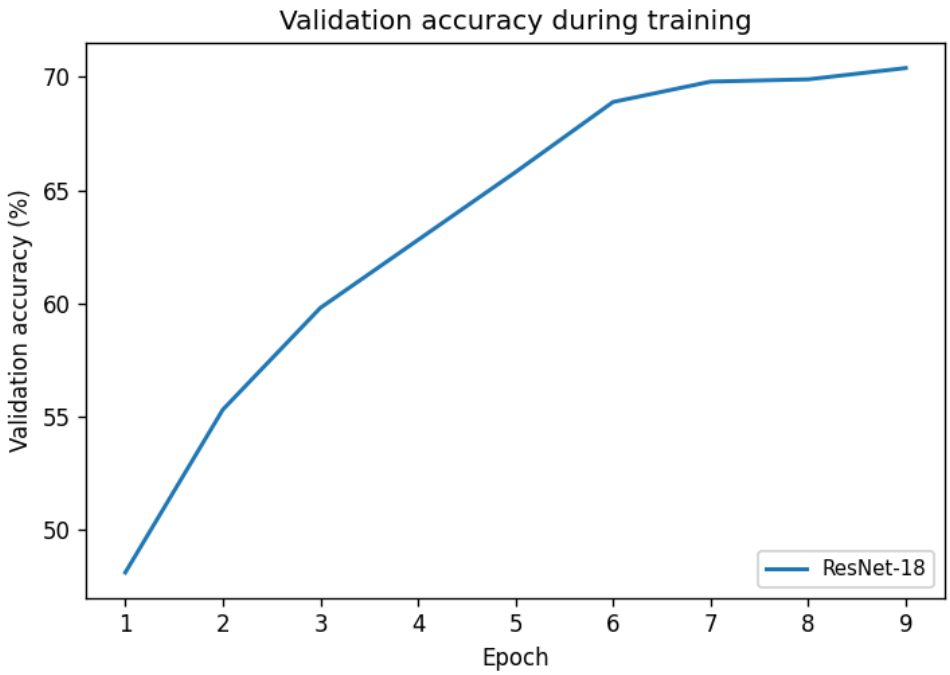


Figure 1: Validation accuracy during training for the models in this experiment.

As shown in Figure 1, ResNet-18 scores 4.6 points higher on 8 than on 4. From epoch 7 to epoch 8, the validation accuracy of ResNet-18 improves by a relative 17.7% (Figure 1). ResNet-18 scores 14.0 points higher on 7 than on 2 (Figure 1). ResNet-18 attains a validation accuracy of 55.3% at epoch 1.

Figure 2 shows the training loss.

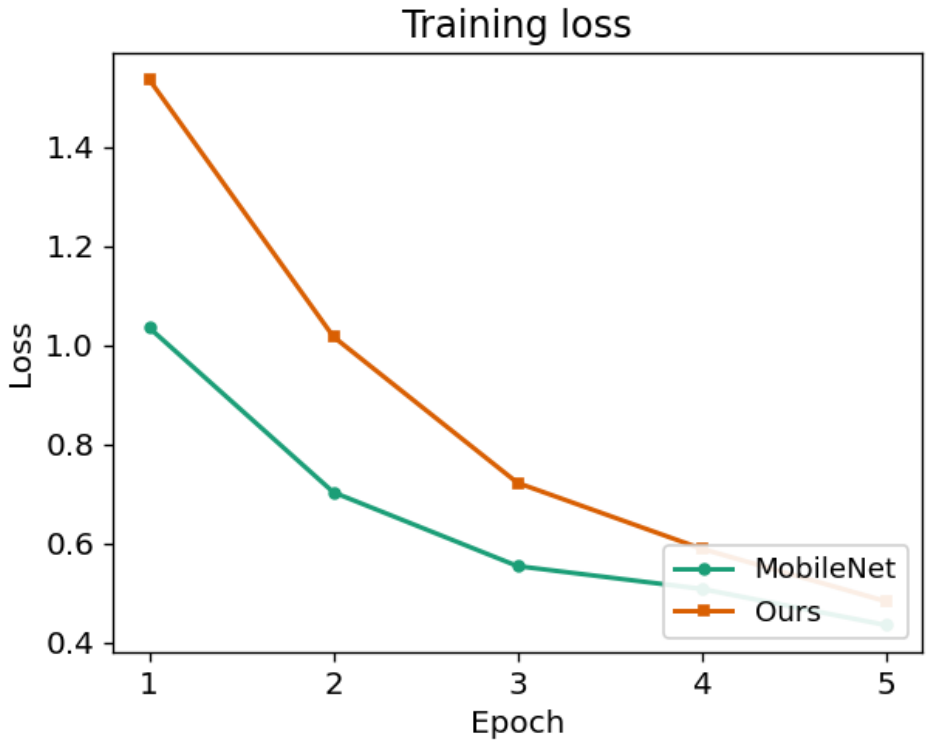


Figure 2: Training loss for the models in this experiment.

Ours has a loss of over 0.9 at epoch 2. MobileNet has a loss of over 1.0 at epoch 1.

Figure 3 shows the validation accuracy by dataset.

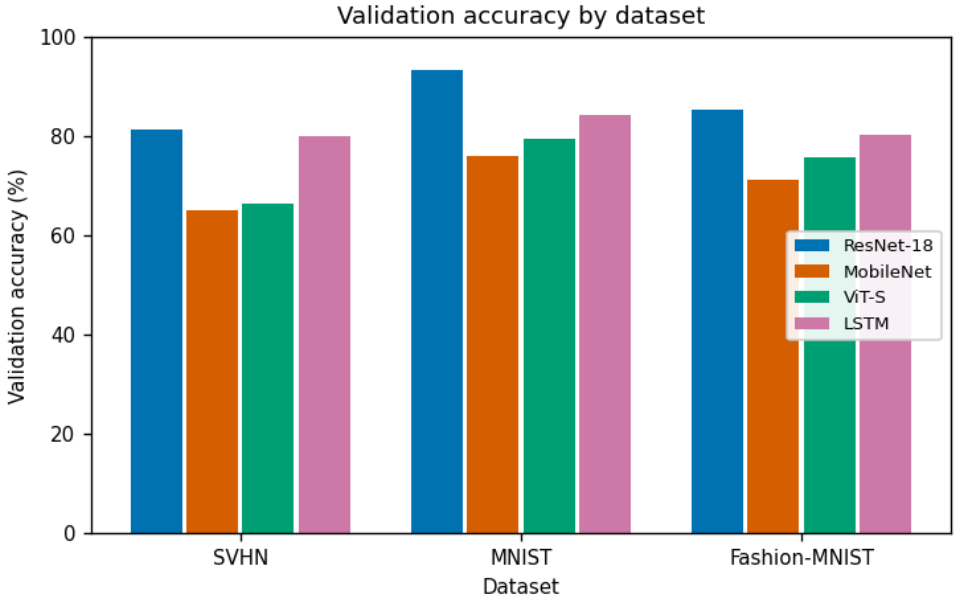


Figure 3: Validation accuracy by dataset for the models in this experiment.

On Fashion-MNIST, ResNet-18 is 19.6 points better than ViT-S. ResNet-18 reaches 85.3% validation accuracy on Fashion-MNIST (Figure 3). ResNet-18 outperforms MobileNet by 66.1 percentage points on SVHN.

Figure 4 shows the training loss.

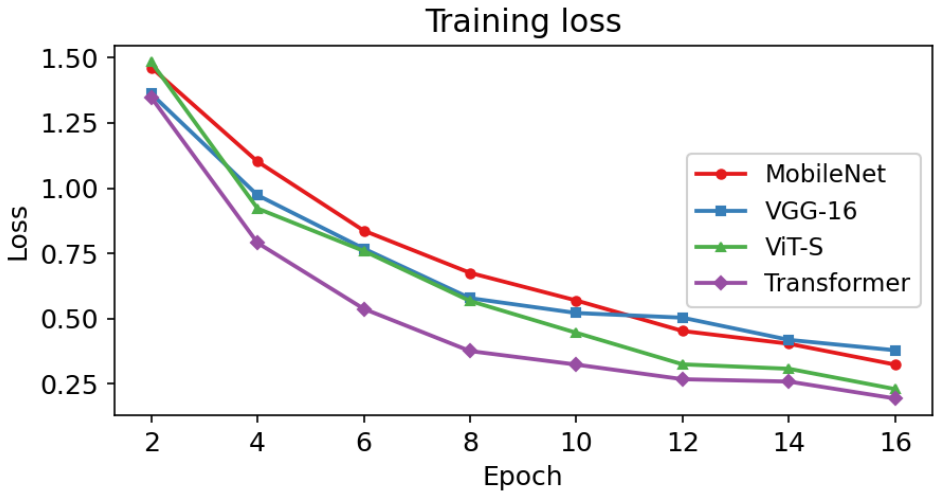


Figure 4: Training loss for the models in this experiment.

As shown in Figure 4, Transformer has a loss of over 0.7 at epoch 4. As shown in Figure 4, at epoch 6, the loss of VGG-16 is 0.768.

3 Discussion

Random Forest reaches 90.8% accuracy on AG News. The differences between the models are discussed in the next section.